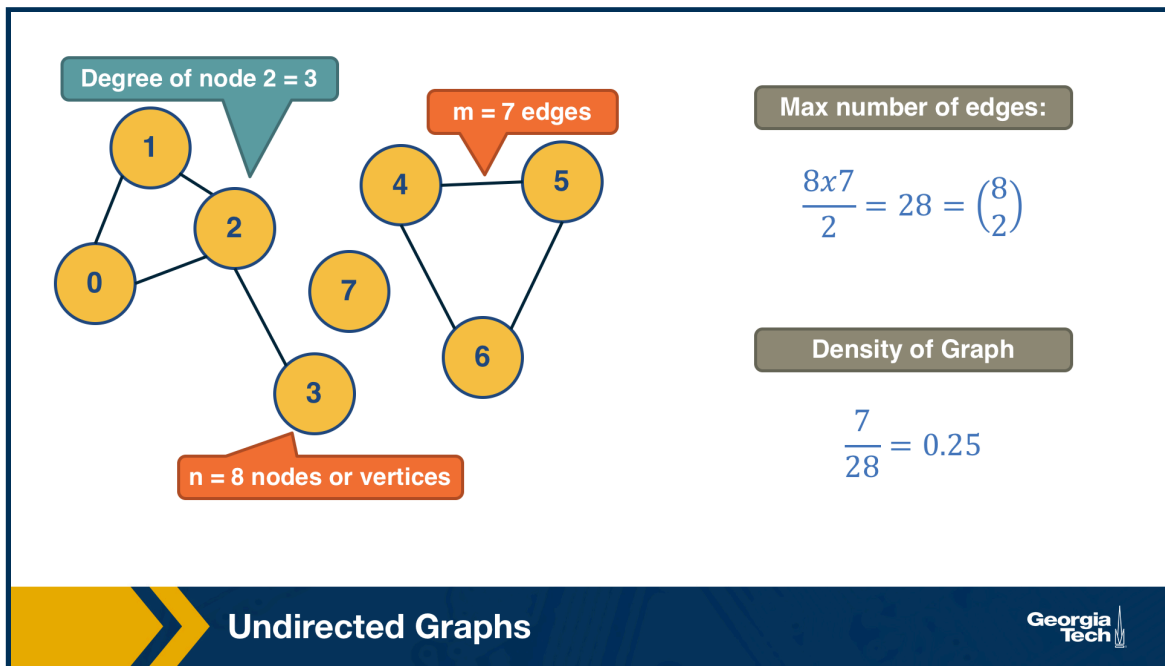


L2: Undirected Graphs

Let's start by defining more precisely what we mean by **graph** or **network** -- we use these two terms interchangeably. We will also define some common types of graphs.



A graph, or network, represents a collection of *dyadic* relations between a set of **nodes**. This set is often denoted by V because nodes are also called **vertices**. The relations are referred to as **edges** or **links**, usually denoted by the set E . So, an edge (u,v) is a member of the set E , and it represents a relation between vertices u and v in the set V .

The number of vertices is often denoted by n and the number of edges by m . We will often use the notation $G=(V,E)$ to refer to a graph with a set of vertices V and a set of edges E . This definition refers to the simplest type of graph, namely *undirected* and *unweighted*.

Typically we do not allow edges between a node and itself. We also do not allow multiple edges between the same pair of nodes. So the maximum number of edges in an undirected graph is $n(n-1)/2$ -- or "n-choose-2". The **density** of a graph is defined as the ratio of the number of edges m by the maximum number of edges (n -choose-2). The number of connections of a node v is referred to as the **degree** of v . The example above illustrates these definitions.